Minimal Functional Dependencies & BCNF Proof

SPICE – Street Food Platform for Information, Customer Reviews & Exploration

Note

The following analysis assumes certain business constraints:

- Each stall has exactly one menu (1:1 relationship)
- Email is unique for each customer
- City_Name is not unique
- A customer can give multiple reviews to the same stall

Under these assumptions, all relations satisfy BCNF.

1. Vendor

Candidate Key(s): {Vendor_ID}

Minimal FD Set:

Vendor_ID → Vendor_Name, Contact_No, Average_Rating, **Years_of_Experience**

BCNF Proof:

Vendor_ID is the only candidate key and a superkey. All non-trivial functional dependencies have a superkey as determinant. All attributes — including **Years_of_Experience** — depend fully on Vendor_ID with no partial or transitive dependency.

Conclusion: Vendor is in BCNF

2. Stall

Candidate Key(s): {Stall_ID}

Minimal FD Set:

Stall_ID → Stall_Name, Location, Opening_Time, Closing_Time, Vendor_ID, City_ID

BCNF Proof:

Stall_ID is the only candidate key. All attributes depend fully on Stall_ID. No subset of attributes determines other attributes; hence no partial or transitive dependency exists.

Conclusion: Stall is in BCNF

3. City

Candidate Key(s): {City_ID}

Minimal FD Set:

City_ID → City_Name, State

BCNF Proof:

City_ID is the only candidate key. City_Name is not assumed to be unique. All non-trivial functional dependencies have superkey determinants.

Conclusion: City is in BCNF

4. Food_Category

Candidate Key(s): {Category_ID}

Minimal FD Set:

Category_ID → Category_Name, Description

BCNF Proof:

Category_ID uniquely determines all attributes. No other non-trivial functional dependency exists.

Conclusion: Food_Category is in BCNF

5. Food_Item

Candidate Key(s): {Food_ID}

Minimal FD Set:

Food_ID → Food_Name, Description, Category_ID

BCNF Proof:

Food_ID is the only candidate key. The only non-trivial functional dependency is Food_ID → Food_Name, Description, Category_ID. Since Food_ID is a superkey, the relation is in BCNF.

Conclusion: Food_Item is in BCNF

6. Menu

Candidate Key(s): {Menu_ID}, {Stall_ID}

Minimal FD Set:

Menu_ID → Stall_ID

Stall_ID → Menu_ID

BCNF Proof:

Since Menu_ID → Stall_ID and Stall_ID → Menu_ID, both attributes determine each other and are candidate keys. Therefore, all non-trivial functional dependencies have superkeys as determinants, and the relation is in BCNF.

Conclusion: Menu is in BCNF

7. Customer

Candidate Key(s): {Customer_ID}, {Email}

Minimal FD Set:

Customer_ID → First_Name, Last_Name, Email, Phone, Registration_Date

Email → Customer_ID (Since Email is unique, it is also a candidate key.)

BCNF Proof:

Email is unique for each customer. Both Customer_ID and Email are candidate keys. All determinants are superkeys.

Conclusion: Customer is in BCNF

8. Review

Candidate Key(s): {Review_ID}

Minimal FD Set:

Review_ID → Rating, Comment, Review_Date, Customer_ID, Stall_ID

BCNF Proof:

Review_ID uniquely identifies each review. A customer can give multiple reviews to the same stall. No other non-trivial functional dependency exists.

Conclusion: Review is in BCNF

9. Authority

Candidate Key(s): {Authority_ID}

Minimal FD Set:

Authority_ID → Name, Department, Contact, Role

BCNF Proof:

Authority_ID uniquely determines all attributes. No other non-trivial functional dependency exists.

Conclusion: Authority is in BCNF

10. License

Candidate Key(s): {License_ID}

Minimal FD Set:

License_ID → Issue_Date, Expiry_Date, Status, License_Type, Vendor_ID, Authority_ID

BCNF Proof:

License_ID is the only candidate key. All attributes depend on License_ID. No other non-trivial functional dependency exists.

Conclusion: License is in BCNF

11. Inspection

Candidate Key(s): {Inspection_ID}

Minimal FD Set:

Inspection_ID → Inspection_Date, Hygiene_Rating, Remarks, Stall_ID, Authority_ID

BCNF Proof:

Inspection_ID uniquely determines all attributes. Multiple inspections per stall are allowed. No other non-trivial functional dependency exists.

Conclusion: Inspection is in BCNF

12. Menu_Item

Relationship: Menu and Food_Item share a **M:N relationship** — a menu can contain multiple food items, and the same food item can appear in multiple menus. Menu_Item is the junction/associative table capturing this many-to-many relationship.

Candidate Key(s): {Menu_ID, Food_ID}

Minimal FD Set:

(Menu_ID, Food_ID) → Price, Availability

BCNF Proof:

The composite key is the only candidate key. The determinant of the functional dependency is a superkey. No other non-trivial functional dependency exists.

Conclusion: Menu_Item is in BCNF

Final Conclusion on BCNF Compliance

All 12 relations satisfy Boyce-Codd Normal Form (BCNF) since for every non-trivial functional dependency, the determinant is a superkey, under the stated business constraints.

Final Minimal FD Set (Fc)

Relation	Minimal FDs (Fc)
Vendor	vendor_id → {vendor_name, contact_no, average_rating, years_of_experience}
Stall	stall_id → {stall_name, location, opening_time, closing_time, vendor_id, city_id}
City	city_id → {city_name, state}
Food_Category	category_id → {category_name, description}
Food_Item	food_id → {food_name, description, category_id}
Menu	menu_id → stall_id ; stall_id → menu_id
Menu_Item (M:N)	(menu_id, food_id) → {price, availability}
Customer	customer_id → {first_name, last_name, email, phone, registration_date} ; email → customer_id
Review	review_id → {rating, comment, review_date, customer_id, stall_id}
Authority	authority_id → {name, department, contact, role}
License	license_id → {issue_date, expiry_date, status, license_type, vendor_id, authority_id}
Inspection	inspection_id → {inspection_date, hygiene_rating, remarks, stall_id, authority_id}

Note: Blue highlighted text indicates updated content.